

Membrane Transport Bench

Three routes across one bilayer. Which ones need ATP, which ones saturate, and why those are two different questions.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/membrane-transport/

1 Before you open anything

Answer from what you already know, then leave it alone. The point is to have a number on paper that the demonstration can later agree or disagree with.

PREDICTION

Glucose enters a cell through carrier proteins in the membrane. The cell starts empty of glucose, and the fluid outside holds 50 mM glucose. You now double the outside concentration to 100 mM. **By what factor does the rate of glucose entry change?**

Write one sentence saying how you got that factor.

2 Reading the bench when it has settled

Open the demonstration. Three lanes run side by side across one membrane, and each has its own control panel on the right.

1. Press **Reset bench**. Every slider returns to its starting value and the model runs.
2. Leave it alone for about **two minutes**, until the readouts stop changing. Do not touch a slider while you wait.
3. Fill in the table from the three route panels. Copy the numbers exactly as displayed, including the words underneath the **Net flux** tiles.

Panel	Readout	What it says
1 · Simple diffusion — O ₂	[O ₂] inside	
	Net flux (and the line under it)	
	Crossing in →	
	Crossing out ←	
2 · Facilitated — glucose	Net flux (and the line under it)	
	Carriers occupied	
	Ceiling V _{max}	
3 · Active transport — Na ⁺	[Na ⁺] inside	
	Gradient the pump is holding	
	Pump rate, out ↑	
	Leak, net in ↓	
	ATP charge	

a. Two lanes finished with a net flux of zero and one did not finish at all. Which lane is still moving solute, and what is it spending to do it?

b. In the O₂ panel, **Net flux** reads zero while **Crossing in →** and **Crossing out ←** are both clearly non-zero and equal. Write one sentence saying what quantity the word *equilibrium* is a statement about.

c. In the Na⁺ panel, **Pump rate, out ↑** and **Leak, net in ↓** read the same number, so [Na⁺] inside is also holding still. Explain how this steady state differs from the one in the O₂ lane, using the ATP charge reading in your answer.

The **Honest limits** panel tells you what the model is holding fixed. The outside is treated as a large bath, so the outside sliders never drift — only the cell interior moves. The cell is one well-mixed compartment of 1 pL, which is why a flux of 1 fmol s^{-1} changes the inside by exactly 1 mM s^{-1} . All three solutes share one 0–200 mM scale so the routes can be compared; real extracellular values are about 145 mM Na^+ , 5 mM glucose and 0.1 mM O_2 .

3 What happens when you keep raising the concentration

Now hold the model still and change one thing at a time. Both passive lanes will be set to an empty cell, which is the same situation the flux chart draws.

1. Press **Pause** in the Bench panel. The button now reads **Run**, and nothing will drift while you work.
2. Drag **[O₂] inside** to **0.0 mM** and **[glucose] inside** to **0.0 mM**.
3. Check that **Carrier proteins** reads **100**. Leave it there for the whole table.
4. Set **[O₂] outside** and **[glucose] outside** both to **20 mM**, and record the two **Net flux** values and **Carriers occupied**.
5. Repeat for each concentration in the table. Keep the chart on the **Flux vs concentration** tab and watch the two curves as you go.

[solute] outside (mM)	O ₂ net flux (fmol s ⁻¹)	Glucose net flux (fmol s ⁻¹)	Carriers occupied
20			
50			
100			
150			
200			

a. Divide the 100 mM flux by the 50 mM flux, once for each route. Give both factors.

O₂: _____ glucose: _____

b. **Ceiling V_{max}** did not move once during the table. What is its value, and what two things in the panel set it?

c. State the rule for each route in your own words: what does raising [outside] do to the flux, and for the route that levels off, name the thing that is running out.

d. Go back to your prediction in part 1. What factor did you write, and what does the table give? If they differ, say specifically what your reasoning assumed that the carrier route does not obey.

A bend in a flux-versus-concentration curve tells you a fixed number of protein binding sites is involved. It tells you nothing at all about energy — part 4 is the test that does.

4 Using it the way a question would

4.1 — Working backwards to a carrier number. Stay paused. Set **[glucose] inside** to **0.0 mM** and **[glucose] outside** to **45 mM**.

a. Using the expression printed in the glucose panel, and $V_{\max} = 0.08 \times \text{carriers}$, work out on paper how many carrier proteins are needed for a net flux of exactly **9.00 fmol s⁻¹**. Show the algebra.

b. Now set the **Carrier proteins** slider to your answer and record what **Net flux** and **Ceiling $V_{\max}$** report.

4.2 — The ATP test. Find the **The ATP test** panel. It reports each route's rate now and the rate the same route would have if the ATP charge were zero, side by side, at this instant.

Route	Rate now	If ATP = 0
Simple — O ₂		
Facilitated — glucose		
Na ⁺ pump		

c. One row differs between the two columns. Which one, and why do the other two rows have to be identical? Answer from what ATP appears in, not from the picture.

d. Press **Reset bench**, then **Cut ATP supply**, and watch the Na⁺ lane for about a minute. Record **[Na⁺] inside**, **ATP charge** and **Gradient the pump is holding** once they stop changing.

4.3 — A shortcut that fails. It is reasonable to expect that a working protein in the membrane can move its solute wherever the cell needs it. Test that on the carrier. Press **Reset bench**, then **Pause**.

1. Set **[glucose] outside** to **20 mM** and **[glucose] inside** to **120.0 mM**.
2. Record **Net flux** and the direction written under it.
3. Raise **Carrier proteins** from 100 to **200** and record both again.

Carrier proteins	Net flux	Direction
100		
200		

e. Adding carriers made the situation worse, not better. Using the numerator of the printed flux expression, explain why no setting of any slider on that panel can make the carrier move glucose into the cell here.

f. In one sentence: what can the Na^+ lane do that neither passive lane can do, no matter how many proteins you give it?

5 Solve these without the demonstration

Close the page or look away from it. Show your working. Then reopen it, set the stated values, and check — and where you were off, write down which step went wrong.

1. A membrane has $P = 0.10 \text{ fmol s}^{-1} \text{ mM}^{-1}$ for O_2 , with 160 mM outside and 40 mM inside. Find the unidirectional flux inward, the unidirectional flux outward, and the net flux. Then raise *both* concentrations by 30 mM and state what happens to each of the three numbers.

2. A membrane carries 75 carrier proteins, each turning over 0.08 fmol s^{-1} at full occupancy, with $K_m = 15 \text{ mM}$. Glucose is 60 mM outside and 10 mM inside. Find $V_{\max}$, the net flux, and the fraction of carriers occupied.

3. An empty cell ($[\text{inside}] = 0$) has both routes available: simple diffusion with $P = 0.10$, and 100 carriers with $K_m = 15 \text{ mM}$. Find the $[\text{outside}]$ at which the two routes carry exactly the same flux. Then predict, with a calculation, where that crossing moves if the carrier count is doubled to 200.

4. A cell is treated with a poison that blocks ATP synthesis completely. Predict what happens over the next few minutes to (i) the rate of O_2 entry, (ii) the rate of glucose entry through its carrier, and (iii) $[Na^+]$ inside. Explain why the presence of a protein in the glucose route does not change your answer, and name what the ATP was buying in the one route that does change.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet uses.

- Switch the chart to **Concentrations over time**, press **Cut ATP supply**, and time how long the Na^+ trace takes to close half the gap to its dashed outside line. Compare with the figure quoted in the caption.
 - Uncheck **Bilayer is permeable to this molecule** in the O_2 panel. That models a charged ion facing bare lipid. Then uncheck **Carrier protein present in the membrane** mid-run and watch where the glucose lane freezes.
 - Set **ATP supply** to 2.0, then 5.0, then 20.0 % s^{-1} , letting each settle. Record the gradient each time. Is the gradient proportional to the supply?
 - Press **Equalise all sides** and watch all three lanes. Only one rebuilds a gradient. Then read the warning at the bottom of **Honest limits**: what real limit on the pump is this model leaving out?
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